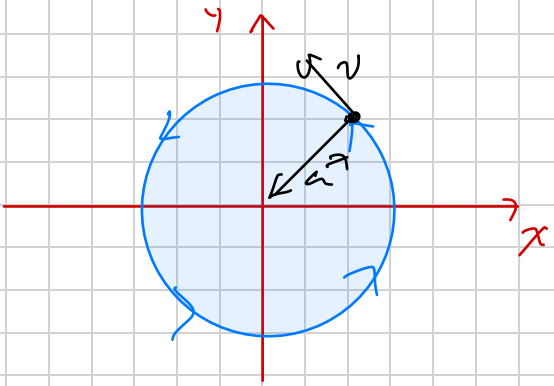
$\vec{r}(t) = (\cos(t), \operatorname{Sen}(t))$

$\vec{r}'(t) = (-\operatorname{Sen}(t), \cos(t))$

$\vec{r}''(t) = (-\cos(t), -\operatorname{Sen}(t))$

$\|\vec{r}'(t)\| = 1$



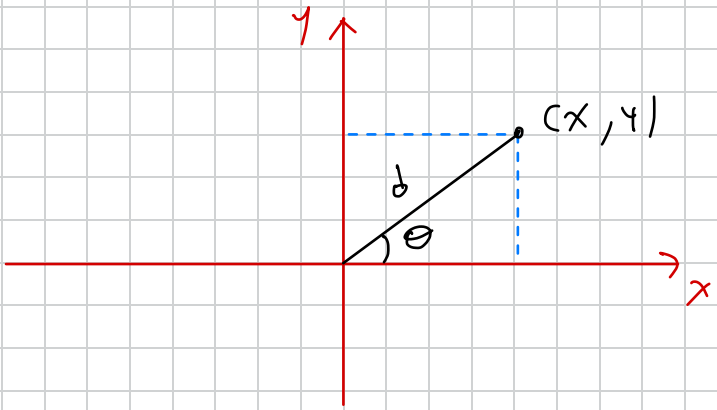
$\vec{r}'(t) \cdot r''(t) = \operatorname{Sen}(t)\cos(t) - \operatorname{Sen}(t)\cos(t) = 0$

$\vec{v} \perp \vec{v}$

Coordenadas cartesianas

Plano xy.

$(x, y) = \{x \in \mathbb{R}, y \in \mathbb{R}\}$



$\operatorname{Sen}\theta = \frac{y}{d}$

$\cos\theta = \frac{x}{d}$

$\tan\theta = \frac{y}{x}$

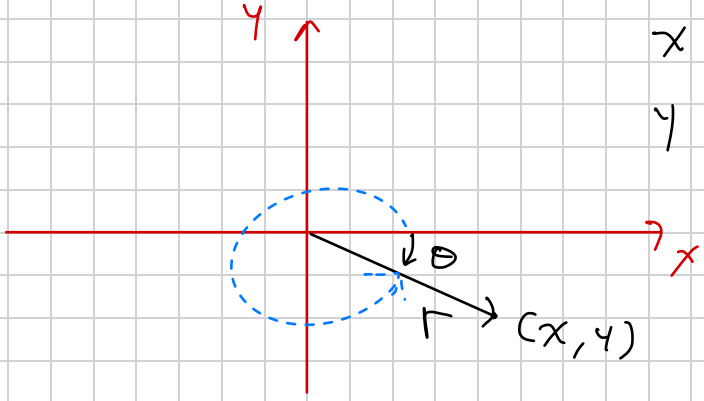
$d^2 = x^2 + y^2$

$d = \sqrt{x^2 + y^2}$

Coordenadas polares: $(x, y) \longrightarrow (r, \theta)$

$r \geq 0 \quad ; \quad 0 \leq \theta \leq 2\pi$

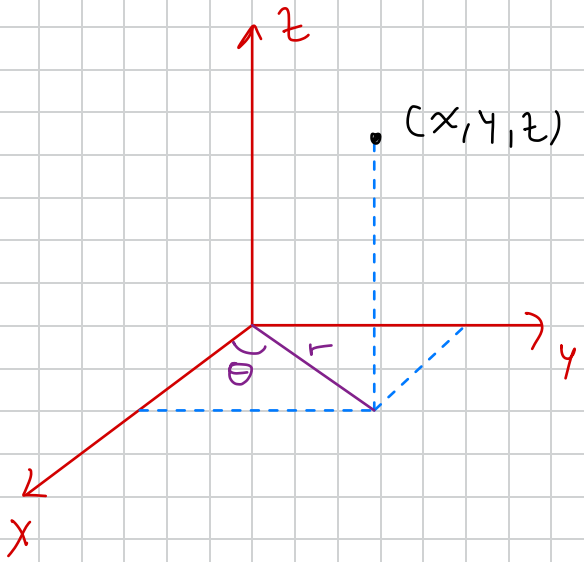
$r = \sqrt{x^2 + y^2} \quad ; \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$



$x = r\cos\theta$

$y = r\operatorname{Sen}\theta$

Coordenadas cilíndricas $(x,y,z) \longrightarrow (r,\theta,z)$



$r = \sqrt{x^2+y^2}$; $r \geq 0$

$0 \leq \theta \leq 2\pi$

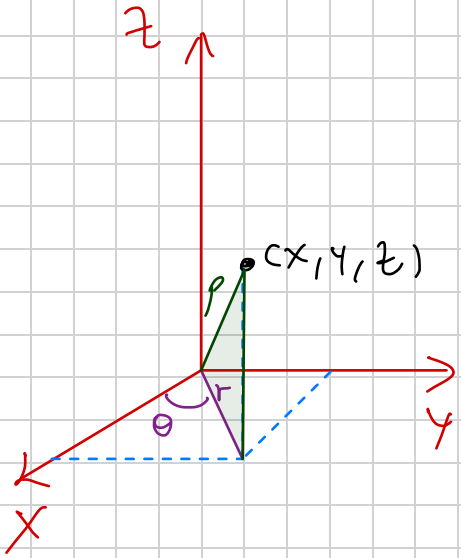
$z \in \mathbb{R}$

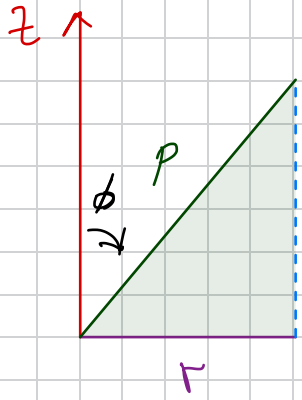
$x = r\cos\theta$

$y = r\operatorname{sen}\theta$

$z = z$

coordenadas esféricas $(x,y,z) \longrightarrow (\rho,\theta,\phi)$





$r = \rho\operatorname{sen}\phi$

$z = \rho\cos\phi$

$x = r\cos\theta = \rho\operatorname{sen}\phi\cos\theta$

$y = r\operatorname{sen}\theta = \rho\operatorname{sen}\phi\operatorname{sen}\theta$

$x = \rho\operatorname{sen}\phi\cos\theta$

$y = \rho\operatorname{sen}\phi\operatorname{sen}\theta$

$z = \rho\cos\phi$

$\rho = \sqrt{x^2+y^2+z^2}$; $\rho \geq 0$

$0 \leq \theta \leq 2\pi$

$0 \leq \phi \leq \pi$